

Toby Nguyen (Nguyễn Đình Tú)

Technical Lead | Senior Software Engineer | Product-Minded Technologist

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Technical Lead & Senior Software Engineer with a **Product-first mindset**, specialized in **distributed backend systems**, **high-scale web applications**, and **optimizing engineering productivity through AI-augmented workflows (MCP, LLMs)**.

CAREER SUMMARY

Accomplished Technical Lead & Senior Software Engineer with over 8+ years of experience engineering high-scale distributed systems, resilient backend services, and world-class web applications. Deeply versatile across systems programming (Rust, Java/Spring Boot, Node.js) and modern web ecosystems (React, Next.js, Vue.js, Micro-frontends). Proven expertise in data pipelines, real-time telemetry streaming (WebSockets, SSE), distributed caching, and Core Web Vitals optimization.

Pioneer in marrying a **Product-Led Growth (PLG)** mindset with **AI-augmented engineering workflows (MCP, LLM toolchains)** to drive technical excellence, deliver resilient software architectures, and scale cross-functional engineering teams.

CORE COMPETENCIES & TECHNICAL SKILLS

Domain	Core Stack & Capabilities
Backend & Systems Engineering	Java (Spring Boot, Security, JPA), Rust (Tokio, Win32 Jobs), Node.js (NestJS, Express), Event-Driven Architecture
API Design & Streaming	REST, GraphQL, gRPC, WebSockets, SSE, API Gateways, BFF, Distributed IPC
Databases & Caching	PostgreSQL, MS SQL Server, Redis (Pub/Sub, Caching), ClickHouse OLAP, Apache Solr, Schema & Index Optimization
Frontend Architecture	React (Next.js, Server Components), Vue.js (Nuxt), TypeScript, Electron JS
Scalability & Performance	Micro-frontends (Module Federation), Design Systems (Storybook, Radix), Core Web Vitals (LCP < 2s, INP < 150ms)
Real-time Data Visualization	Canvas/SVG Charting, Virtual Grids, ECharts, Highcharts, TanStack Query, SWR
DevOps & Cloud	Docker, Kubernetes fundamentals, Azure Cloud, Linux CLI, GitHub Actions CI/CD, Turborepo Monorepo

Domain	Core Stack & Capabilities
AI-Augmented Engineering	Model Context Protocol (MCP), LLM Toolchains, Playwright, Vitest, Jest
Engineering Leadership	Technical Strategy, 15+ Engineers Leadership, Architecture Roadmaps, Tech Debt Governance, Agile Scrum

PROFESSIONAL EXPERIENCE

Company	Role	Duration
CMC Global	Technical Project Lead	09/2024 – Present
ICOMM Tech	CPO & Tech Lead	05/2020 – 08/2024
OpenCommerce Group	Senior Software Engineer	12/2018 – 04/2020
TDT Asia	Full-stack Engineer	05/2018 – 12/2018
Lush Era	Software Engineer	12/2016 – 04/2018

FEATURED PROJECT LIST

[09/2024 – Present] High-Scale Web & Booking Portals – Theme Park Enterprise

- **Timeline:** 09/2024 – Present
- **Company:** CMC GLOBAL
- **Client:** Multinational Theme Park & Hospitality Enterprise (South Korea)
- **Project description:** High-scale digital ecosystem for premier international amusement parks, encompassing high-traffic web portals & partner booking webviews, real-time attraction wait-time & reservation engines, and digital interactive park navigation.
- **Team size:** 15+ engineers
- **Responsibilities:**
 - Served as **Lead Frontend Engineer**, steering the architectural refactoring, component modularity, and feature delivery for high-traffic Park Visitor Web Portals and Partner Booking WebViews adhering to client enterprise standards.
 - Implemented responsive Web and mobile-embedded WebView user interfaces using React, Vite, and Tailwind CSS, ensuring UI fidelity with design documentation and smooth interactions.

- Configured streamlined local developer workflows and mock data synchronization between Spring Boot backend services and React / React Native frontend clients.
- Performed continuous code maintenance, bug fixes, and performance tuning to guarantee cross-browser compatibility and responsive UX across mobile WebViews.
- **Programming Language:** TypeScript, JavaScript, Java
- **Framework:** React, Vite, React Native, Tailwind CSS, Spring Boot
- **Platform, server and database:** Docker, Redis, MS SQL Server, Linux
- **Methodology:** Agile Scrum

[09/2024 – Present] Digital Twin – Electric Vehicle (EV) Real-Time Telemetry Dashboard

- **Timeline:** 09/2024 – Present
- **Company:** CMC GLOBAL
- **Client:** Leading Automotive Manufacturer (Automotive Tech, IoT Integration)
- **Team size:** 15+ engineers (3 squads)
- **Project description:** Large-scale real-time Digital Twin telemetry dashboard platform for aggregating, monitoring, and analyzing telemetry and operational status for fleets of electric vehicles (EVs).
- **Responsibilities:**
 - Served as **Technical Project Lead**, establishing overall technical architecture, engineering standards, and real-time telemetry streaming across 3 squads (15+ engineers) processing millions of daily EV operational events.
 - Architected a **Micro-frontends** architecture and standardized a shared enterprise **Design System**, ensuring visual consistency and decoupled release cycles across squads.
 - Optimized real-time telemetry streaming via WebSockets by implementing **Virtual Scrolling, Canvas Data Charting, and Debounced/Throttled State Updates**, while orchestrating high-throughput BFF aggregation services in **Rust** to sustain a continuous 60fps rendering rate under high telemetry volume without blocking the Main Thread.
 - Formulated a comprehensive **Core Web Vitals** strategy: orchestrated route-based and component-based dynamic code splitting, reducing INP to < 150ms and maintaining LCP < 2.0s across all mission-critical monitor views.
 - Introduced **AI-augmented Frontend SDLC** practices (LLM UI generation, automated visual regression testing), elevating frontend feature delivery velocity by 25%.
- **Programming Language:** TypeScript, JavaScript, HTML5, CSS3, Rust (BFF tooling)
- **Framework:** React (Concurrent Features), Next.js, Tailwind CSS, Zustand, Node.js
- **Platform, server and database:** Azure Cloud, Docker, Kubernetes, WebSockets, PostgreSQL, Redis
- **Methodology:** Agile Scrum

[05/2020 – 08/2024] Big Data Analytics & Interactive Intelligence Platform

- **Timeline:** 05/2020 – 08/2024
- **Company:** ICOMM TECH.JSC
- **Client:** Government, Banking & Aviation Sectors

- **Team size:** 15+ engineers
- **Project description:** Multi-platform enterprise Big Data extraction and intelligence dashboard, delivering interactive data visualization, real-time reporting, and deep analytics under strict enterprise security constraints.
- **Responsibilities:**
 - Directed product roadmap and technical architecture as **Chief Product Officer (CPO) & Technical Lead**.
 - Engineered an interactive **Data Visualization Dashboard** powered by ECharts and Highcharts, solving high-density data rendering challenges for 100,000+ complex visual data points without frame drops.
 - Implemented **Client-side Caching (SWR/React Query patterns)** and **Virtualized Data Grids**, enabling near-instantaneous search, filtering, and paging across millions of records.
 - Established automated frontend quality assurance suites (End-to-End testing with Cypress/Playwright and unit testing with Jest), achieving 99.9% application uptime SLA.
 - Mentored 15+ engineers in Component-Driven Architecture, JavaScript Clean Code principles, and runtime render profiling.
- **Programming Language:** TypeScript, JavaScript, HTML5, SCSS/CSS3, Python
- **Framework:** React Ecosystem, Redux Toolkit, ECharts, .NET Core (Backend API)
- **Platform, server and database:** Docker, Kubernetes, Redis, Apache Solr, ClickHouse, Linux
- **Methodology:** Agile Scrum

[12/2018 – 04/2020] High-Converting E-Commerce Storefront & Merchant Suite

- **Timeline:** 12/2018 – 04/2020
- **Company:** OPEN COMMERCE GROUP
- **Client:** Global E-commerce Merchants
- **Team size:** 8–10 members
- **Project description:** Ultra-fast e-commerce storefront platform and intelligent cross-selling engine integrated across Shopify, WooCommerce, and WordPress ecosystems.
- **Responsibilities:**
 - Served as **Senior Software Engineer**, spearheading storefront user interface development and AI-driven cross-selling recommendation widgets geared toward Product-Led Growth (PLG).
 - Maximized conversion rate optimization (CRO) and checkout velocity: drove **First Contentful Paint (FCP) to < 1.2s** and **LCP to < 1.8s**, contributing to an estimated 15–20% increase in merchant revenue.
 - Engineered reusable, modular UI components using **Vue.js and React**, enforcing scoped styles, responsive image optimization (WebP/AVIF), and aggressive bundle size minimization.
 - Delivered seamless Mobile-First responsive layouts and frictionless multi-gateway payment flows.
- **Programming Language:** TypeScript, JavaScript, HTML5, CSS3, Rust (microservices)
- **Framework:** Vue.js (Vuex), React, Tailwind CSS, Laravel (Backend API)
- **Platform, server and database:** AWS, Nginx, Redis, MySQL
- **Methodology:** Agile Scrum

[05/2018 – 12/2018] Enterprise Japanese Business Management Portal

- **Timeline:** 05/2018 – 12/2018

- **Company:** TDT ASIA JSC
- **Client:** Japanese Small & Medium Enterprises (SMB Solutions – Japan Market)
- **Team size:** 6–8 members
- **Project description:** Enterprise Resource Planning (ERP) and business management web portal customized to meet the rigorous quality and UX standards of the Japanese corporate market.
- **Responsibilities:**
 - Operated as **Fullstack Software Engineer**, taking primary ownership of the Single Page Application (SPA) client architecture.
 - Leveraged **Angular with RxJS Observables** for reactive state handling and orchestrated complex business workflows with Angular Reactive Forms.
 - Ensured strict **Cross-Browser Compatibility** (Chrome, Firefox, Safari, Edge) and achieved pixel-perfect adherence to Japanese UX/UI specifications.
- **Programming Language:** TypeScript, JavaScript, HTML5, SCSS
- **Framework:** Angular, Angular Material, RxJS, Spring Boot (Backend API)
- **Platform, server and database:** Tomcat, Docker, MySQL, Linux
- **Methodology:** Agile Scrum

[12/2016 – 04/2018] Responsive Real-time Telemedicine Web Application

- **Timeline:** 12/2016 – 04/2018
- **Company:** LUSH ERA LLC
- **Client:** Medical Clinics & Healthcare Service Providers
- **Team size:** 5 members
- **Project description:** Secure real-time telemedicine and remote healthcare consultation web application connecting patients with certified medical specialists via video and chat.
- **Responsibilities:**
 - Acted as **Software Engineer**, building an intuitive, accessible, and fast-loading telemedicine web client.
 - Engineered **Mobile-First Responsive Layouts**, prioritizing Web Accessibility (a11y) standards for elderly patients and individuals with visual impairments.
 - Implemented interactive WebRTC video calling interfaces and low-latency real-time messaging UI.
- **Programming Language:** JavaScript, HTML5, CSS3, PHP
- **Framework:** jQuery, Bootstrap, Laravel (Backend)
- **Platform, server and database:** MySQL, Apache, Ubuntu Linux
- **Methodology:** Agile

[2024 – Present] Tuquet – Developer CLI & Distributed Automation Ecosystem

- **Timeline:** 2024 – Present
- **Company:** Independent Product Labs (Open-Source R&D)
- **Client:** Self-initiated Open-Source Developer Tooling & Global Automation Engineers
- **Team size:** Principal Architect (Solo Project)

- **Project description:** High-performance open-source developer tooling ecosystem and distributed browser automation suite consisting of a unified master CLI (`tuquet`) in Rust, a visual workflow studio & cross-platform desktop suite (Vue.js / Nuxt, Electron JS), a supervised background runner daemon (`tuquet runner`), and cloud coordination services.
- **Responsibilities:**
 - Acted as **Creator & Principal Architect**, architecting the end-to-end multi-runtime ecosystem spanning systems programming in Rust, visual web studio in Vue.js, and distributed task supervision.
 - Engineered the master CLI (`tuquet`) in **Rust**: built an interactive scoped shell with hierarchical state management, rustyline-based smart tab-completion, ANSI color styling, and sub-command delegation (`automa` , `runner` , `cloud` , `browser`).
 - Developed the visual automation studio & cross-platform desktop suite with **Vue.js (Nuxt), Electron JS & TypeScript**: designed an interactive node-graph canvas (Drawflow), custom automation blocks, secure local IPC bridges, and a real-time execution inspector with instant visual feedback.
 - Architected a zero-leakage process supervision engine in the runner daemon (`tuquet runner`) leveraging Win32 Job Objects to eliminate orphan Chromium zombie processes and strictly isolate CPU/RAM.
 - Engineered a high-throughput Medallion data crawler pipeline (Bronze gzip BLOB storage -> Silver Rust deduplication & HTML sanitizer -> Gold Supabase synchronization).
 - Standardized local workspace configurations into a Single Source of Truth (`~/ .tuquet/`), eliminating configuration drift across CLI, extensions, and runner runtimes.
 - Automated binary builds, multi-platform packaging, and distribution through an official Windows Scoop bucket (`tuquet/scoop-bucket`).
 - Published modular open-source ecosystem libraries to the official NPM registry under the `@tuquet` scope (`@tuquet/md-export` , `@tuquet/lunar` , `@tuquet/vue-ui` , `@tuquet/extension-runner`).
- **Programming Language:** Rust, TypeScript, JavaScript, SQL
- **Framework:** Vue.js Ecosystem (Nuxt, Drawflow), Electron JS, Tailwind CSS, Tokio (Rust Async), Node.js
- **Platform, server and database:** Supabase, PostgreSQL, WebSockets, Win32 Job Objects, Docker, Scoop, GitHub Actions
- **Methodology:** Agile Kanban

ACHIEVEMENTS & AWARDS

- **Q4/2025: Rising Star Award (CMC GLOBAL)** – Recognized for exemplary technical leadership and high-impact delivery of the Digital Twin EV project.
- **2023: Team of the Year (ICOMM TECH.JSC)** – Leading the engineering organization to exceed all product delivery targets.
- **2022: Outstanding Employee (ICOMM TECH.JSC)** – Awarded for spearheading the flagship Big Data platform.

EDUCATION

FPT University | Information Technology (1/2012 – 9/2014)

- **Focus:** Full-Stack Web Development, Desktop Application Design & Software Architecture.
- **Academic Standing:** GPA 8.2/10
- **English Proficiency:** TOEIC 650 (Fluent technical reading/writing & professional collaboration)